LEXIGRID

CLI and Privileged Administration

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Procedures, examples, verification and operational limits. Complete commissioning on your deployment before serving customer traffic. Sample values are fictional; original keys and data must be preserved.

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CLI and privileged administration

Command families and safe context

Run commands from the verified installation directory. ./lexictl --version reads VERSION; ./lexictl --help and subcommand help describe implemented options. Root lifecycle wrappers support deployment-mode behavior; scripts/lexictl.py supplies native status/logs, migration inspection, AI diagnostics and owner-administration dispatch.

The native CLI loads .env.native as its default runtime configuration. To select a separate maintenance connection, inject LEXIGRID_ADMIN_DATABASE_URL into the operator's environment through approved secret storage. This explicit override applies only to lexictl db-current/db-history/db-upgrade and admin commands. An explicitly empty value blocks the operation; it never falls back silently to the installed database. The setting is not persisted to .env.native or passed to AI/lifecycle children.

Do not put the URL/password on the command line, in shell history or in a support ticket. A shell DATABASE_URL alone still does not override .env.native for lexictl; use the dedicated maintenance setting. Direct python -m alembic / app.admin_cli commands use their separately configured DATABASE_URL. Keep the API's installed connection restricted; do not replace it with an owner connection merely for maintenance.

Privileged app.admin_cli operations verify the actual connected PostgreSQL role owns public.users (or is a controlled superuser), and always reject lexigrid_runtime before requesting a password or changing data. URL encoding/aliases do not bypass the check. SQLite filenames are not interpreted as PostgreSQL role names. A failed connection/role check produces a generic error without echoing the URL. These checks do not replace explicit maintenance authorization and a verified backup.

Read-only diagnostics

```
./lexictl --version
./lexictl --help
./lexictl status
./lexictl health
./lexictl db-current
./lexictl db-history
./lexictl ai providers
./lexictl ai status
```

Health identifies the expected native instance; an unrelated listener must not count as success. Migration inspection still imports application configuration and needs a valid session secret. AI status can show operational configuration; treat output as restricted and never dump it indiscriminately into public issues.

./lexictl logs --lines 60 displays recent native log entries. Review locally for sensitive details before sharing. AI models/test can contact the configured provider and therefore are not offline-safe merely because they are diagnostic commands. ai configure prints configuration guidance; it is not a browser secret editor.

Native lifecycle

```
./install.sh --help
./start.sh
./status.sh
./stop.sh
```

Use native manager help for explicit native database/host/port options. Installer selection occurs before Docker prerequisites. The installer preserves existing configuration and keys; do not delete .env.native to clear a startup problem. Native mode does not manage PostgreSQL or create a system supervisor/scheduler automatically.

Before an approved restart with a migration, verify the final consistent backup and compatible restored upgrade. Verify complete shutdown before snapshotting or reusing the port. The manager refuses an unrelated instance; investigate a failed identity check rather than force-killing an arbitrary process.

Provision a tenant

From an approved owner-configured environment:

```
LEXIGRID_ADMIN_CONFIRM=YES python -m app.admin_cli create-tenant \
  --name 'Example organization' --email admin@example.invalid
```

The CLI prompts privately for a password of at least 16 characters. Example addresses are fictional. Verify the named tenant and initial administrator in the intended database, the audit event and a real designated test login. Duplicate organization names are rejected; do not reset the database to retry provisioning.

Credential recovery

```
LEXIGRID_ADMIN_CONFIRM=YES python -m app.admin_cli reset-password \
  --tenant-id TENANT_UUID --email user@example.invalid
```

Enter the replacement password interactively. The operation increments the token version to revoke prior sessions and records a privileged recovery event. This is not an MFA-seed read or automatic MFA reset workflow. Preserve the existing MFA encryption key and investigate enrollment separately.

Bind Entra/OIDC or AD FS/SAML

```
LEXIGRID_ADMIN_CONFIRM=YES python -m app.admin_cli bind-oidc \
  --tenant-id TENANT_UUID --email user@example.invalid \
  --issuer https://login.microsoftonline.com/DIRECTORY_UUID/v2.0 \
  --subject VERIFIED_APPLICATION_SUBJECT

LEXIGRID_ADMIN_CONFIRM=YES python -m app.admin_cli bind-saml \
  --tenant-id TENANT_UUID --email user@example.invalid \
  --subject VERIFIED_STABLE_NAMEID
```

Bind only independently verified IdP identities to an existing account. OIDC sub is application-specific; do not substitute email/UPN/object ID. Binding revokes previous application sessions. Duplicate identity bindings are rejected. See the federation manual for exact callback/MFA context configuration and tests.

Database upgrade

In the schema-owner environment with the correct original secrets and a verified restore path, direct Alembic commands remain available. Alternatively, inject LEXIGRID_ADMIN_DATABASE_URL and use lexictl db-current/db-upgrade; confirm the intended target first. No maintenance command changes the installed runtime connection:

```
python -m alembic current
python -m alembic heads
python -m alembic upgrade head
```

Use runtime credentials only for the API/workers, not migration operations. Stop or coordinate traffic/workers as required by the maintenance plan. Duplicate reminder schedules must be investigated before the 0018 uniqueness index can be applied. Do not remove audit or delivery history by unsupported downgrade.

Independent scheduler and audit jobs

```
python -m app.workflow_scheduler --once
python -m app.audit_anchor --help
python -m app.audit_anchor create --tenant-id TENANT_UUID --output CHECKPOINT_PATH
```

Scheduler once performs work and may send configured notifications; it is not a read-only status command. Avoid running it against production without the intended worker authorization/topology. Continuous scheduling must be supervised separately in native mode.

Restricted PostgreSQL checkpoint jobs require explicit tenant scope. WORM publication reads LEXIGRID_AUDIT_TENANT_ID and independent storage credentials; its success must include retention/read-back verification. Keep anchor keys and output files private.

Uninstallation

Use ./uninstall.sh only after consistent backups and identification of the actual installation path. Normal removal preserves local configuration/log/data to a sibling preservation location and does not delete external PostgreSQL or Docker volumes. Direct native-manager uninstall removes its virtual environment and retains configuration/data; it is not identical to the root wrapper's application-directory cleanup. See the retention manual before using any destructive flag.

Administrator attribution and password rotation

Create the initial deployment administrator with `admin create-tenant --role platform_admin`; the default role for subsequent organization provisioning is tenant_admin. Owner CLI audit events identify the process operating-system account and database role separately from the affected application account. These attributes identify the operating context, not a verified human login. Preserve private operator access records alongside exported audit history.

Users change local passwords in Security with their current password and a fresh authenticator code when MFA is enabled. All prior sessions are revoked. Identity-provider passwords must be changed at the provider.

Dedicated PostgreSQL identities

Database identities must be explicit and separate from an operator's operating-system login. Native installation defaults to `lexigrid_owner` for privileged setup, and the application uses restricted `lexigrid_runtime` credentials after provisioning. A DBA may choose a dedicated maintenance account such as `lexiadmin` using `--pg-user lexiadmin`; create it under the deployment's approved database administration procedure. Migration-created roles use fixed product names, never the current OS username. The installer does not automatically create a personal or privileged account named after the person running it. SQLite uses a database file and has no database login.

For a supplied PostgreSQL URL, put the intended account explicitly before @, for example `postgresql+psycopg://lexigrid_runtime@db.example.invalid/lexigrid` (supply credentials through the approved configuration or secret mechanism). URLs such as `postgresql:///lexigrid` and URLs with `user` / `username` query overrides are rejected before a connection is attempted. Installation, application startup, maintenance, runtime provisioning, diagnostics and the backup script apply the same check. The operating-system username, PGUSER and database-service defaults cannot supply an omitted identity.

Before upgrading an existing PostgreSQL deployment with an implicit username URL, have the DBA supply the approved dedicated account explicitly. Do not rename roles, reset passwords, replace encryption keys or recreate databases simply to satisfy this setting. Existing explicit connections, original credentials and data are retained. Audit records may still identify the actual OS operator separately from the database role.